light stars becoming white dwarfs when they die, and heavy stars becoming neutron stars, there would be no way for nature to make a black hole. Einstein and Eddington, and most physicists and astronomers of their era, would heave a sigh of relief.

Zwicky had been lured to Caltech in 1925 by Millikan. Millikan expected him to do theoretical research on the quantum mechanical structures of atoms and crystals, but more and more during the late 1920s and early 1930s, Zwicky was drawn to astrophysics. It was hard not to be entranced by the astronomical Universe when one worked in Pasadena, the home not only of Caltech but also of the Mount Wilson Observatory, which had the world's largest telescope, a reflector 2.5 meters (100 inches) in diameter.

In 1931 Zwicky latched on to Walter Baade, a new arrival at Mount Wilson from Hamburg and Göttingen and a superb observational astronomer. Baade and Zwicky shared a common cultural background: Baade was German, Zwicky was Swiss, and both spoke German as their native language. They also shared respect for each other's brilliance. But there the commonality ended. Baade's temperament was different from Zwicky's. He was reserved, proud, hard to get to know, universally well informed—and tolerant of his colleagues' peculiarities. Zwicky would test Baade's tolerance during the coming years until finally, during World War II, he and Zwicky would split violently. "Zwicky called Baade a Nazi, which he wasn't, and Baade said he was afraid that Zwicky would kill him. They became a dangerous pair to put in the same room," recalls Jesse Greenstein.

During 1932 and 1933, Baade and Zwicky were often seen in Pasadena, animatedly conversing in German about stars called "novae," which suddenly flare up and shine 10,000 times more brightly than before; and then, after about a month, slowly dim down to normalcy. Baade, with his encyclopedic knowledge of astronomy, was aware of tentative evidence that, in addition to these "ordinary" novae, there might also be unusual, rare, superluminous novae. Astronomers at first had not suspected that those novae were superluminous, since they appeared through telescopes to be roughly the same brightness as ordinary novae. However, they resided in peculiar nebulas (shining "clouds"); and in the 1920s, observations at Mount Wilson and elsewhere began to convince astronomers that those